Week 7, Lecture 14 - Gaussian Mixture Models

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Outline

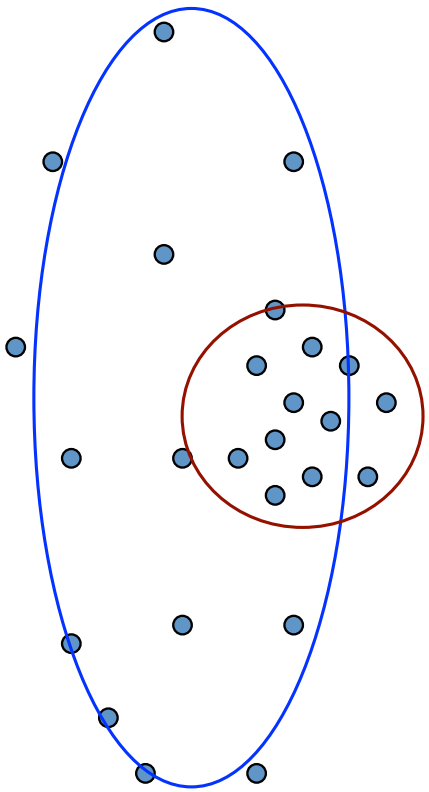
- Administrative Issues
- Gaussian mixtures
- Implementation

Based on slides from David Sontag.

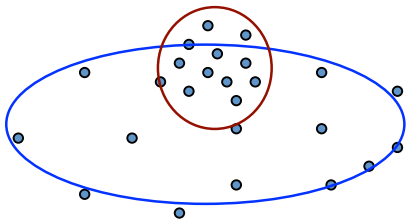
Gaussian mixtures

The Evils of "Hard Assignments"?

- Clusters may overlap
- Some clusters may be "wider" than others
- Distances can be deceiving!



Probabilistic Clustering



- Try a probabilistic model!
  - Allows overlaps, clusters of different size, etc.
- Can tell a *generative story* for data
  - $P(X \mid Y)P(Y)$
- Challenge: we need to estimate model parameters without labeled Ys

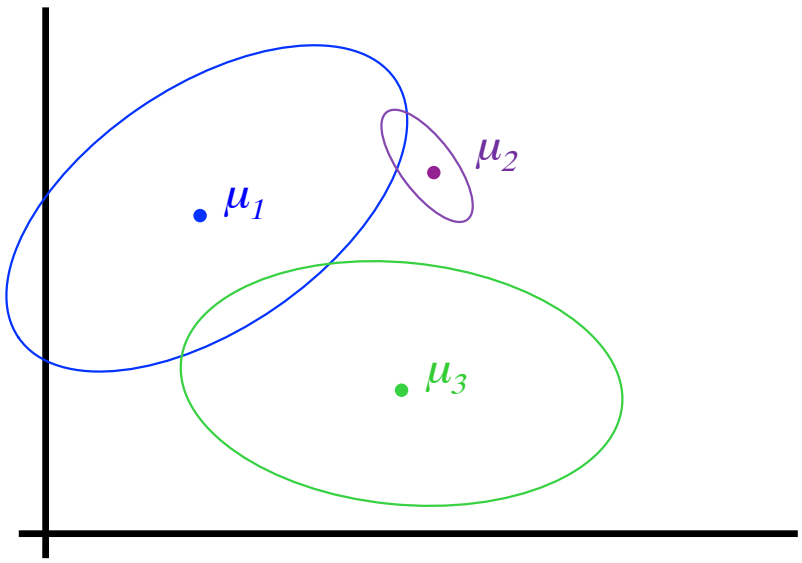
Y	X_1	X_2
??	0.1	2.1
??	0.5	-1.1
??	0.0	3.0
??	-0.1	-2.0
??	0.2	1.5
...	...	...

The General GMM assumption

- $P(Y)$: There are k components
- $P(X \mid Y)$: Each component generates data from a *multivariate Gaussian* with mean μ_i and covariance matrix Σ_i

Each data point is sampled from a **generative process**:

1. Choose component i with probability $P(y = i)$
2. Generate datapoint $N(m_i, \Sigma_i)$



What Model Should We Use?

- Depends on X.
- If we know which points are in a cluster, then we can define the best distribution for it.
  - Multinomial over clusters Y
  - (Independent) Gaussian for each X_i given Y

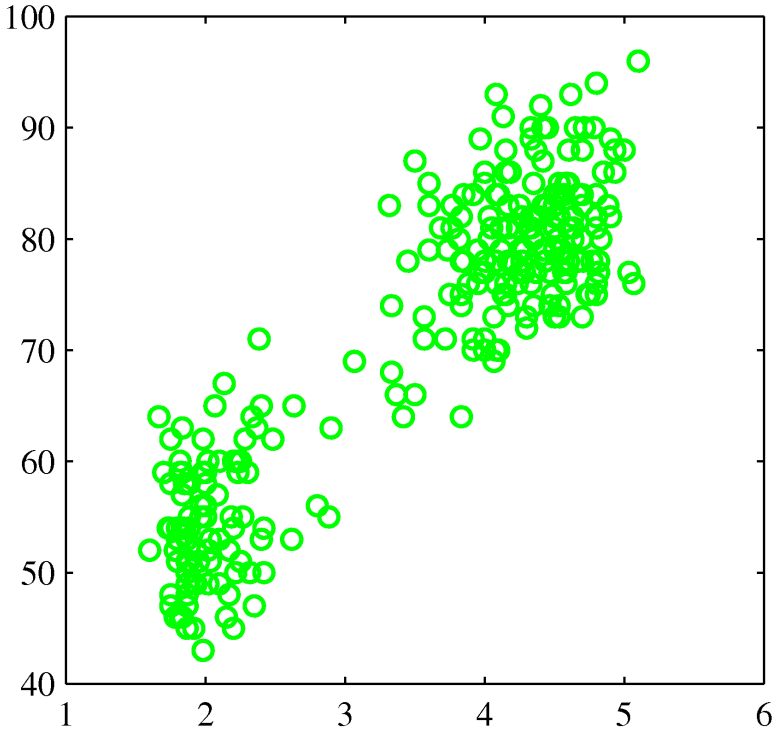
$$p(Y_i = y_k) = \theta_k$$

$$P(X_i = x \mid Y = y_k) = N(x \mid \mu_{ik}, \sigma_{ik})$$

Could we make fewer assumptions?

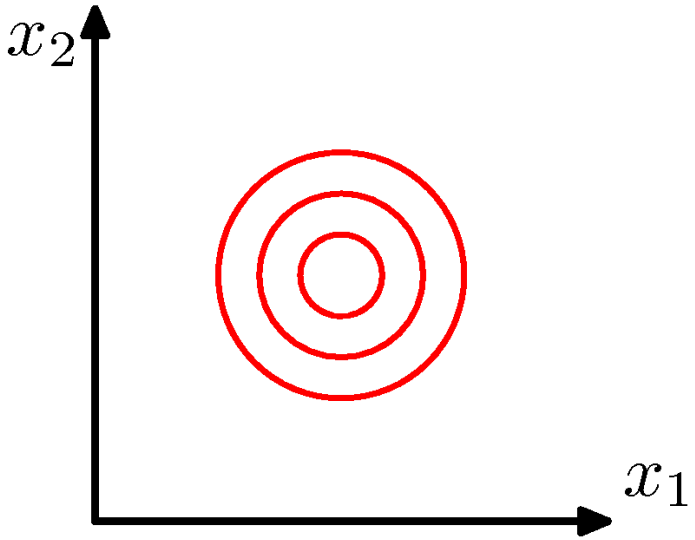
- What if the X_i co-vary?
- What if there are multiple peaks?
- **Gaussian Mixture Models!**
  - $P(Y)$ still multinormal
  - $P(\mathbf{X} \mid Y)$ is a *multivariate* Gaussian distribution:

$$P(X = x_j \mid Y = i) = N(x_j, \mu_i, \Sigma_i)$$



Multivariate Gaussians

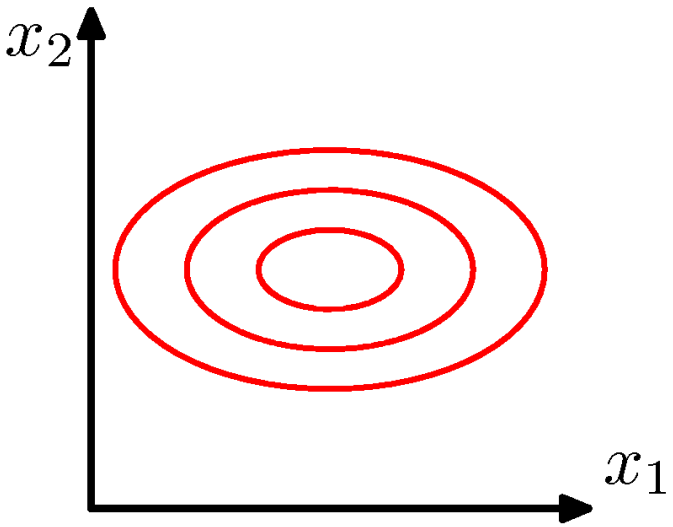
$$P(X=\mathbf{x}_j) = \frac{1}{(2\pi)^{m/2}\,\|\Sigma\|^{1/2}} \exp\left[-\frac{1}{2}\left(\mathbf{x}_j - \mu\right)^T \Sigma^{-1}\left(\mathbf{x}_j - \mu\right)\right]$$



$\Sigma \propto$ identity matrix

Multivariate Gaussians

$$P(X=\mathbf{x}_j) = \frac{1}{(2\pi)^{m/2}\,\|\Sigma\|^{1/2}} \exp\left[-\frac{1}{2}\left(\mathbf{x}_j - \mu\right)^T \Sigma^{-1}\left(\mathbf{x}_j - \mu\right)\right]$$

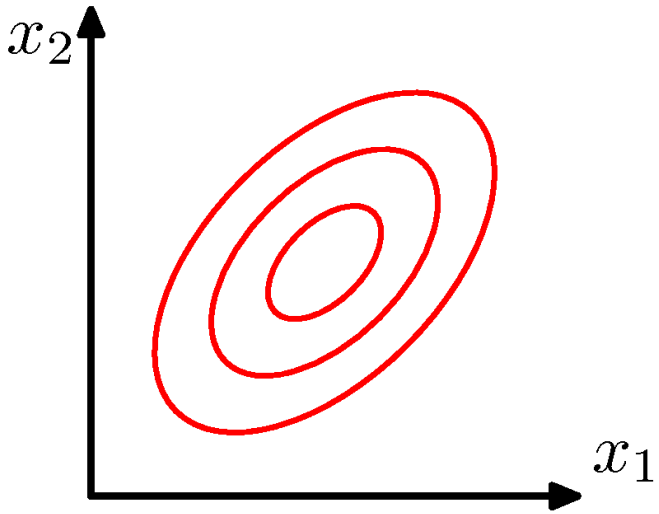


Σ = diagonal matrix

X_i are independent *ala* Gaussian NB

Multivariate Gaussians

$$P(X=\mathbf{x}_j) = \frac{1}{(2\pi)^{m/2}\,\|\Sigma\|^{1/2}} \exp\left[-\frac{1}{2}\left(\mathbf{x}_j - \mu\right)^T \Sigma^{-1}\left(\mathbf{x}_j - \mu\right)\right]$$

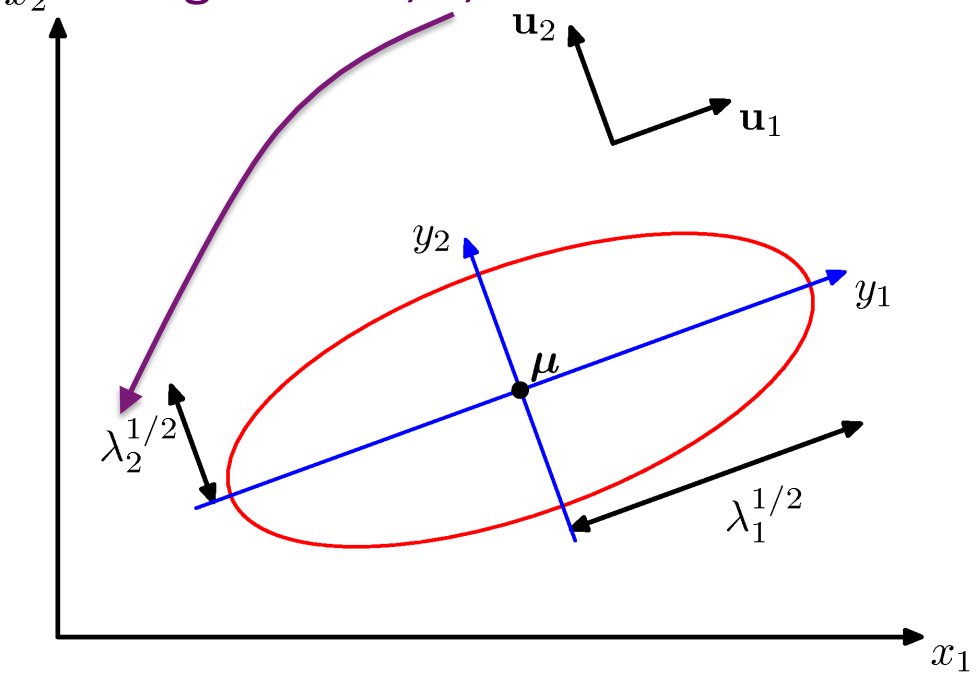


Σ = arbitrary (semidefinite) matrix:

- specifies rotation (change of basis)
- eigenvalues specify relative elongation

Multivariate Gaussians

Eigenvalue, λ, of Σ

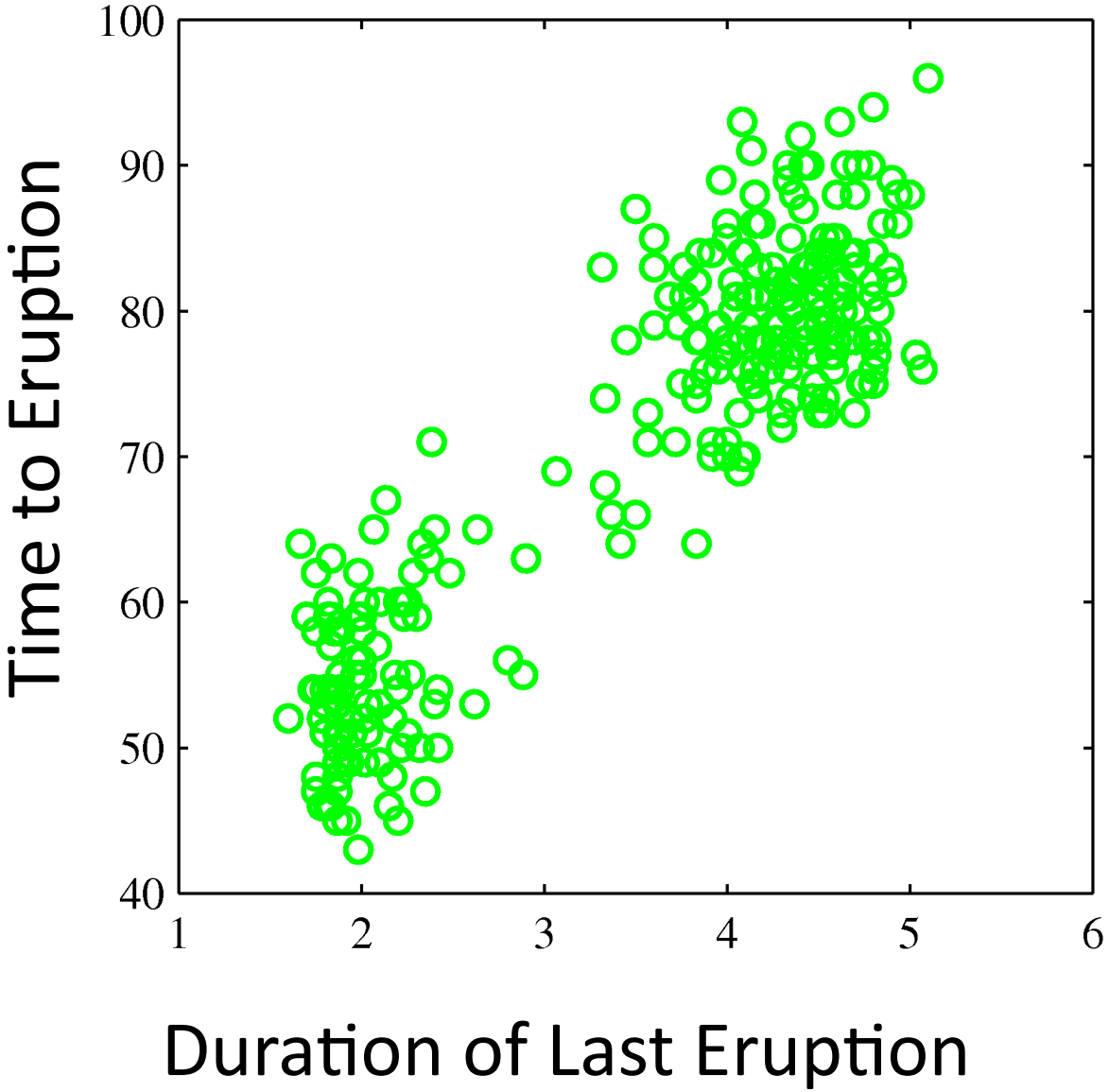


Covariance matrix, Σ, = degree to which x_i vary together

$$P(X=\boldsymbol{x}_j) = \frac{1}{(2\pi)^{m/2}\,\|\Sigma\|^{1/2}} \exp\left[-\frac{1}{2}\left(\mathbf{x}_j-\boldsymbol{\mu}\right)^T \Sigma^{-1}\left(\mathbf{x}_j-\boldsymbol{\mu}\right)\right]$$

Mixtures of Gaussians (1)

Old Faithful Data Set



Mixtures of Gaussians (1)

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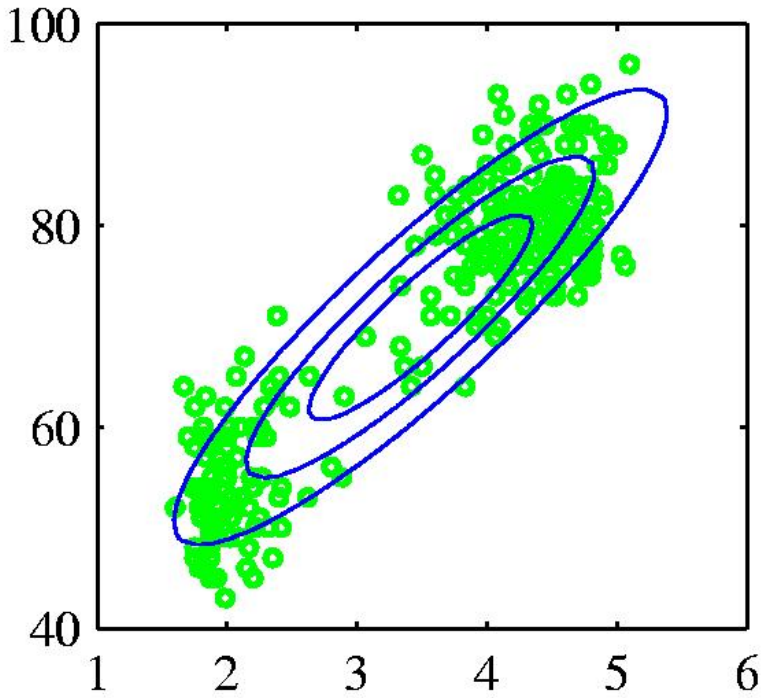
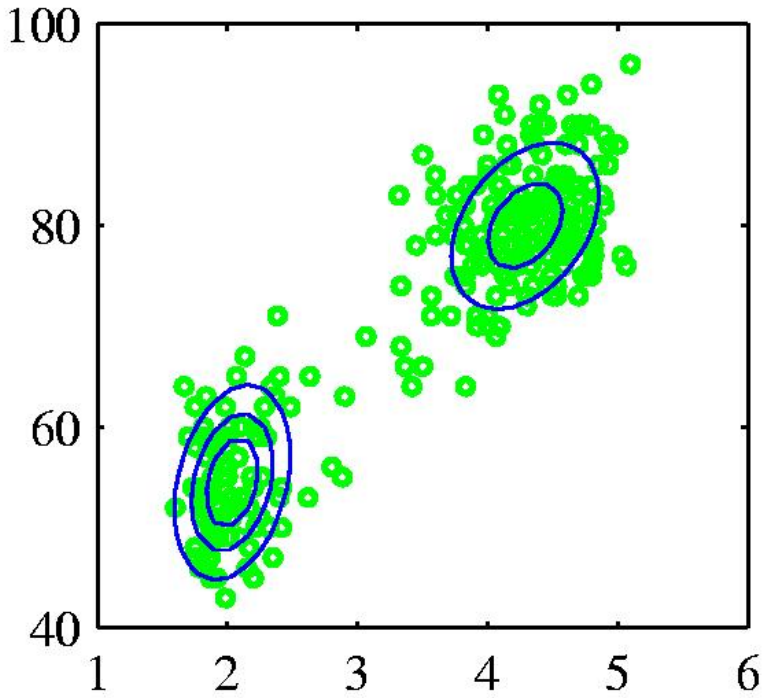


Single Gaussian



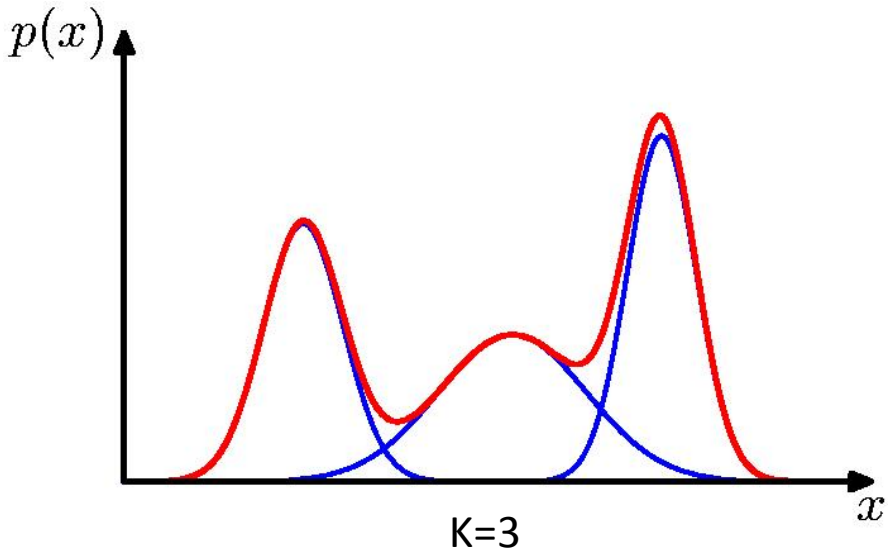
Mixture of two Gaussians

Mixtures of Gaussians (2)

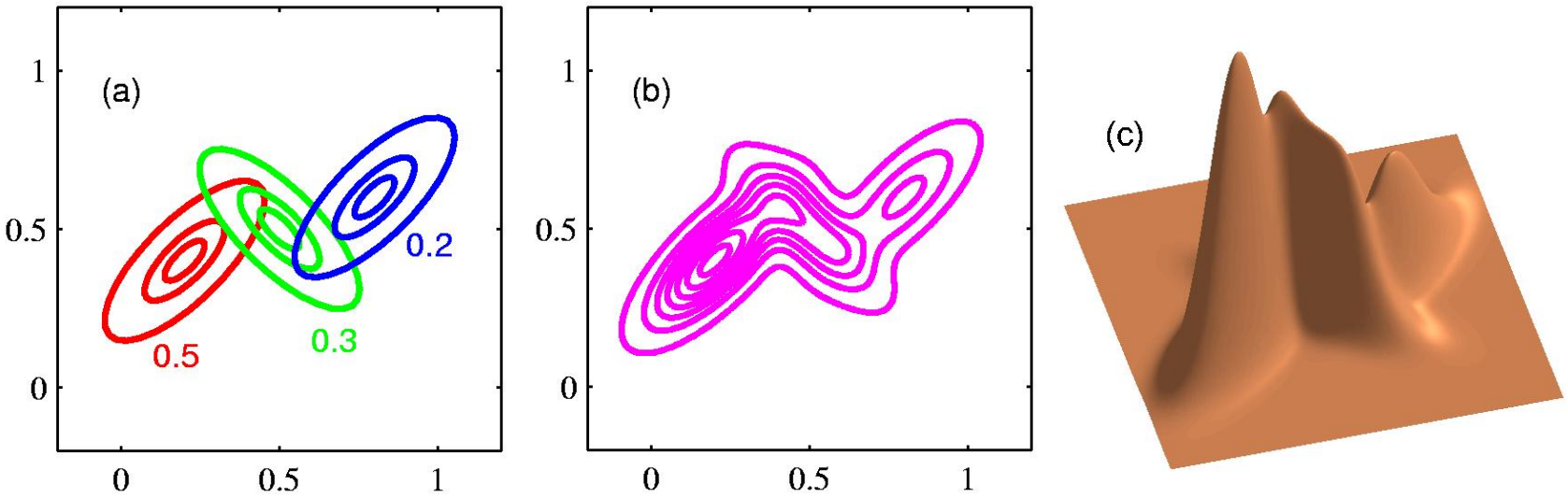
Combine simple models into a complex model:

$$p(\mathbf{x}) = \sum_{k=1}^{K} \underbrace{\pi_k}_{\text{Mixing coefficient}} \underbrace{\mathcal{N}(\mathbf{x}|\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)}_{\text{Component}}$$

$$\forall k : \pi_k \geqslant 0 \qquad \sum_{k=1}^{K} \pi_k = 1$$

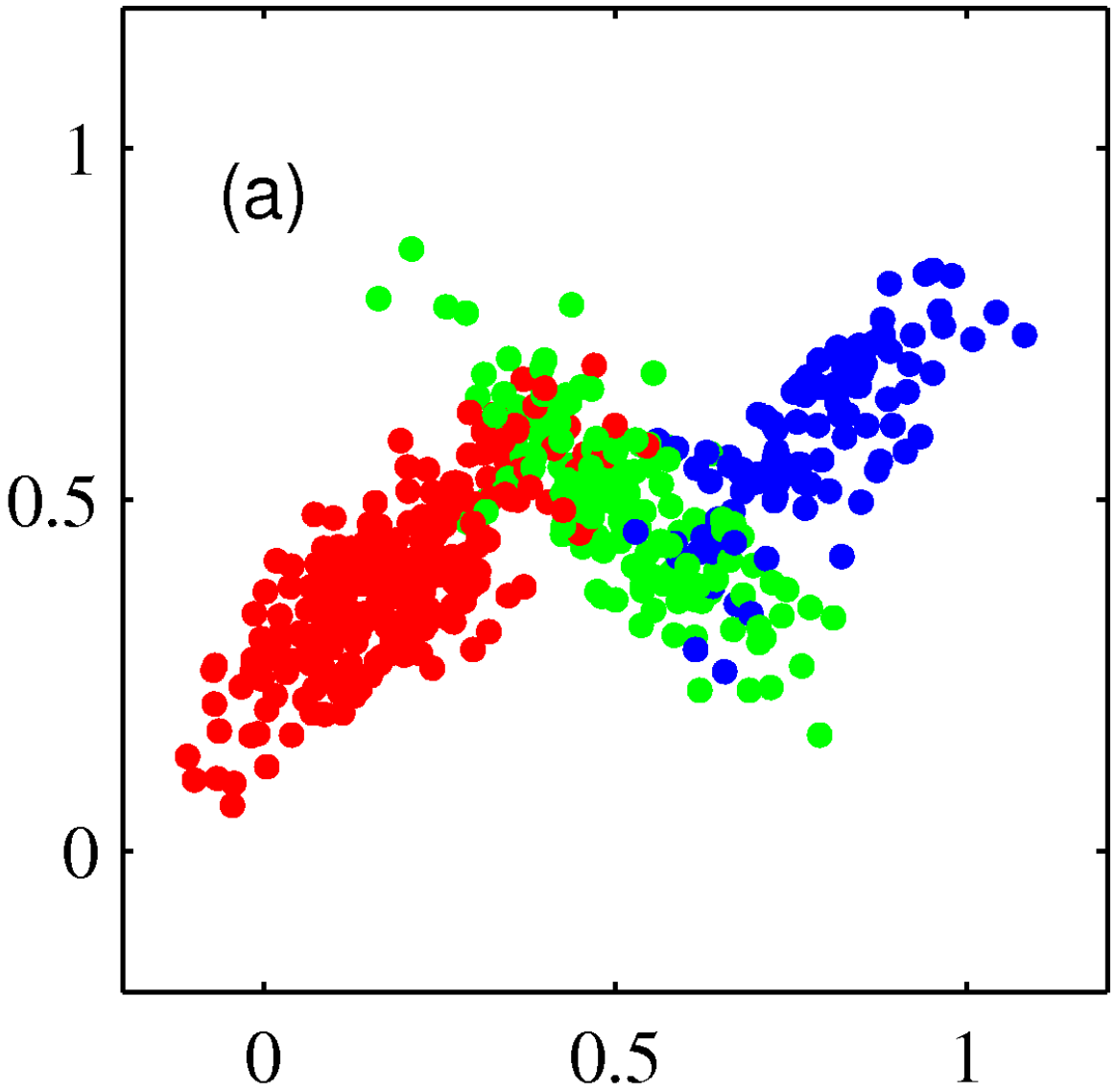


Mixtures of Gaussians (3)



Eliminating Hard Assignments to Clusters

Model data as mixture of multivariate Gaussians



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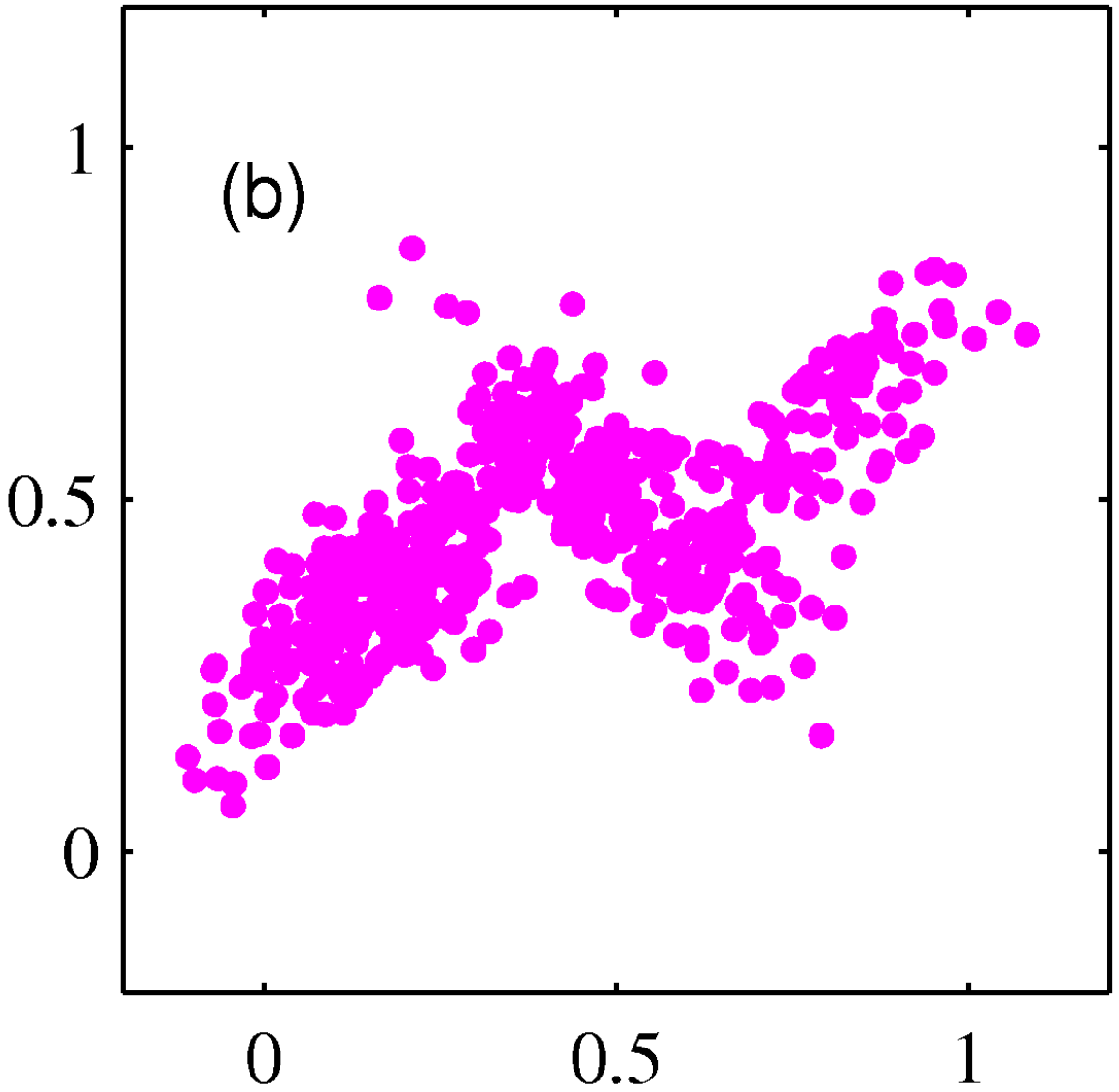
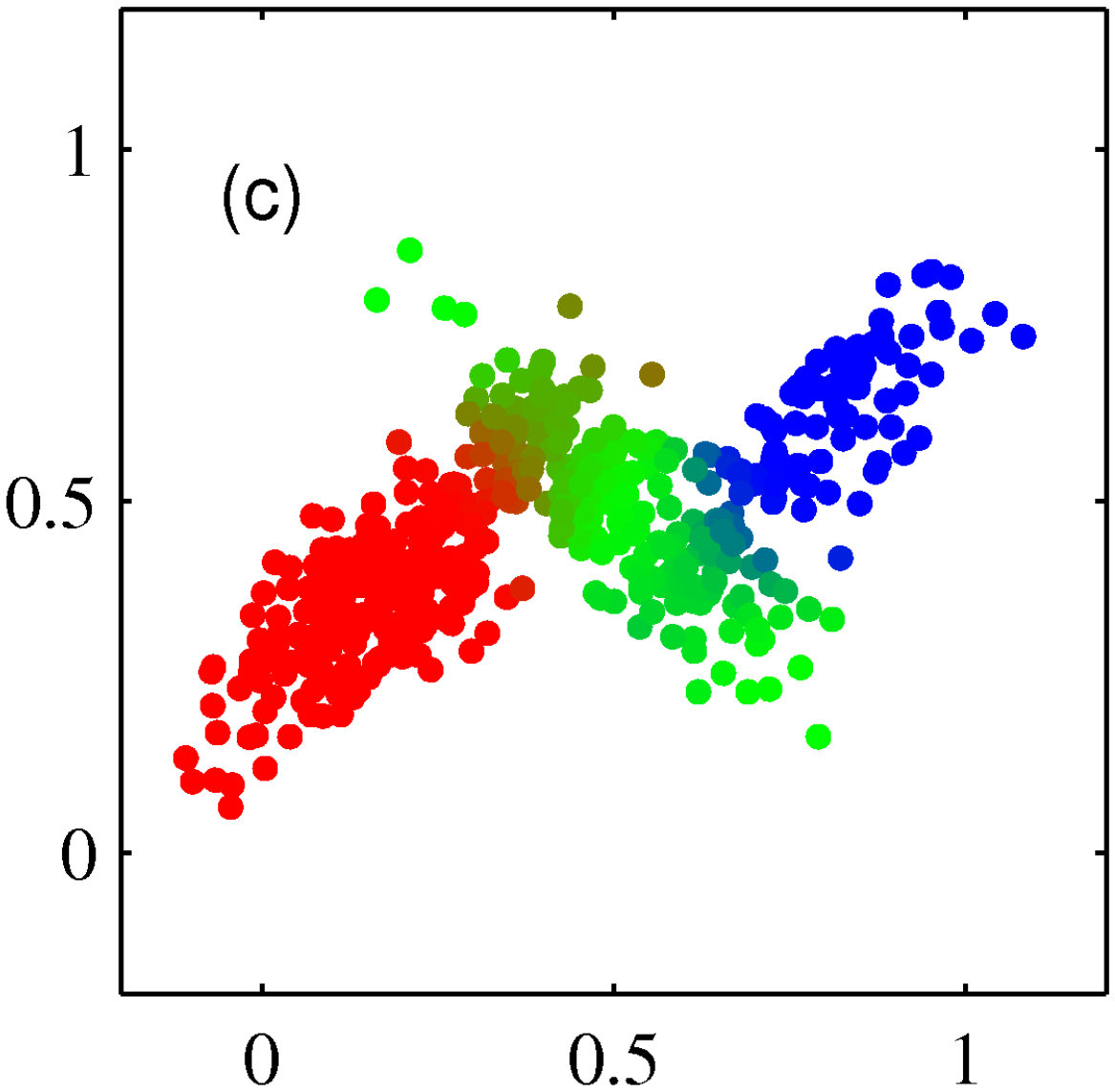


Eliminating Hard Assignments to Clusters

Model data as mixture of multivariate Gaussians



ML estimation in **supervised** setting

- Univariate Gaussian

$$\mu_{MLE} = \frac{1}{N}\sum_{i=1}^{N} x_i \qquad \sigma^2_{MLE} = \frac{1}{N}\sum_{i=1}^{N}(x_i - \hat{\mu})^2$$

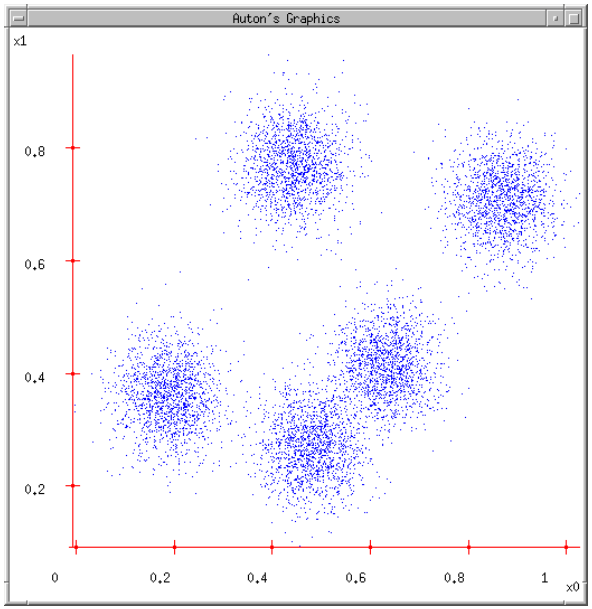
- **Mixture** of **Multi**variate Gaussians
  - ML estimate for each of the Multivariate Gaussians is given by:

$$\mu^k_{ML} = \frac{1}{n}\sum_{j=1}^{n} x_n \qquad \Sigma^k_{ML} = \frac{1}{n}\sum_{j=1}^{n}\left(\mathbf{x}_j - \mu^k_{ML}\right)\left(\mathbf{x}_j - \mu^k_{ML}\right)^T$$

Just sums over x generated from the k'th Gaussian

But what if unobserved data?

- MLE:
  - $\arg\max_\theta \prod_j P(y_i, x_j)$
  - θ: all model parameters
    - eg, class probs, means, and variances
- But we don't know y_j's!
- Maximize **marginal likelihood**:
  - $\arg\max_\theta \prod_j P(x_j) = \arg\max_\theta \prod_j \sum_{k=1}^K P(Y_j = k, x_j)$



How do we optimize? Closed Form?

- Maximize **marginal likelihood**:

$$\arg\max_{\theta} \prod_{j} P(x_j) = \arg\max \prod_{j} \sum_{k=1}^{K} P(Y_j = k, x_j)$$

- Almost always a hard problem!
  - Usually no closed form solution
  - Even when lgP(X,Y) is convex, lgP(X) generally isn't...
  - For all but the simplest P(X), we will have to do gradient ascent, in a big messy space with lots of local optima...

Learning general mixtures of Gaussians

$$P(y=k \mid \mathbf{x}_j) \propto \frac{1}{(2\pi)^{m/2} \|\Sigma_k\|^{1/2}} \exp\left[-\frac{1}{2}\left(\mathbf{x}_j - \mu_k\right)^T \Sigma_k^{-1}\left(\mathbf{x}_j - \mu_k\right)\right] P(y=k)$$

- Marginal likelihood:

$$\prod_{j=1}^{m} P(\mathbf{x}_j) = \prod_{j=1}^{m} \sum_{k=1}^{K} P(\mathbf{x}_j, y=k)$$

$$= \prod_{j=1}^{m} \sum_{k=1}^{K} \frac{1}{(2\pi)^{m/2} \|\Sigma_k\|^{1/2}} \exp\left[-\frac{1}{2}\left(\mathbf{x}_j - \mu_k\right)^T \Sigma_k^{-1}\left(\mathbf{x}_j - \mu_k\right)\right] P(y=k)$$

- Need to differentiate and solve for μ_k, $\sum_k$, and P(Y=k) for k=1..K
- There will be no closed form solution, gradient is complex, lots of local optimum
- **Wouldn't it be nice if there was a better way!?!**

EM

Expectation Maximization

The EM Algorithm

- A clever method for maximizing marginal likelihood:
  - $\arg\max_\theta \prod_j P(x_j) = \arg\max_\theta \prod_j \sum_{k=1}^K P(Y_j = k, x_j)$
  - A type of gradient ascent that can be easy to implement
    - e.g. no line search, learning rates, etc.
- Alternate between two steps:
  - Compute an expectation
  - Compute a maximization
- Not magic: **still optimizing a non-convex function with lots of local optima**
  - The computations are just easier (often, significantly so!)

EM: Two Easy Steps

Objective: $\text{argmax}_\theta \lg \prod_j \sum_{k=1}^K P(Y_j=k, x_j \mid \theta) = \sum_j \lg \sum_{k=1}^K P(Y_j=k, x_j \mid \theta)$

Notation a bit inconsistent
Parameters = $\theta=\lambda$

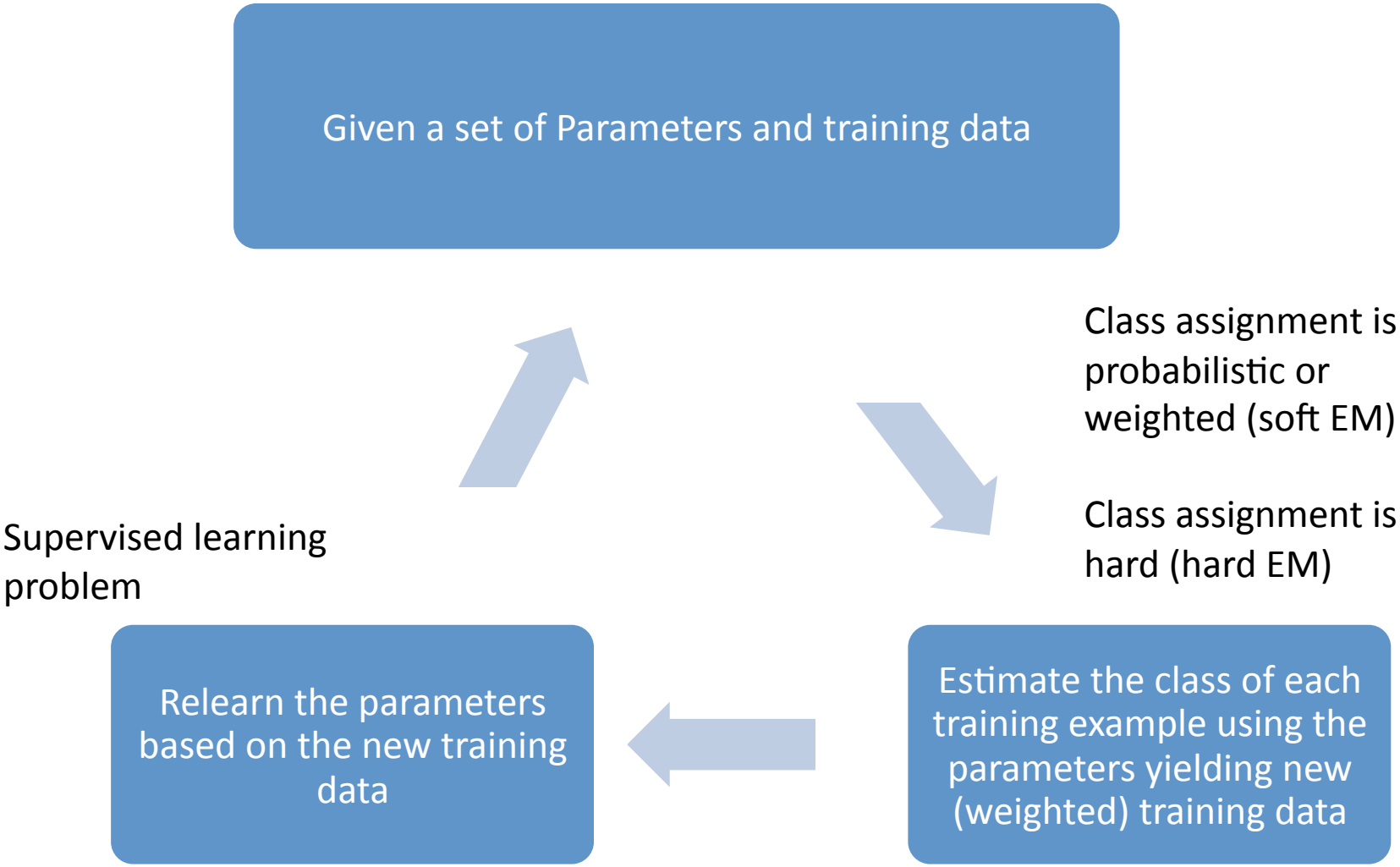
Data: $\{x_j \mid j=1 .. n\}$

- **E-step**: Compute expectations to "fill in" missing y values according to current parameters, θ
  - For all examples j and values k for Y_j, compute: $P(Y_j=k \mid x_j, \theta)$

- **M-step**: Re-estimate the parameters with "weighted" MLE estimates
  - Set $\theta = \text{argmax}_\theta \sum_j \sum_k P(Y_j=k \mid x_j, \theta) \log P(Y_j=k, x_j \mid \theta)$

Especially useful when the E and M steps have closed form solutions!!!

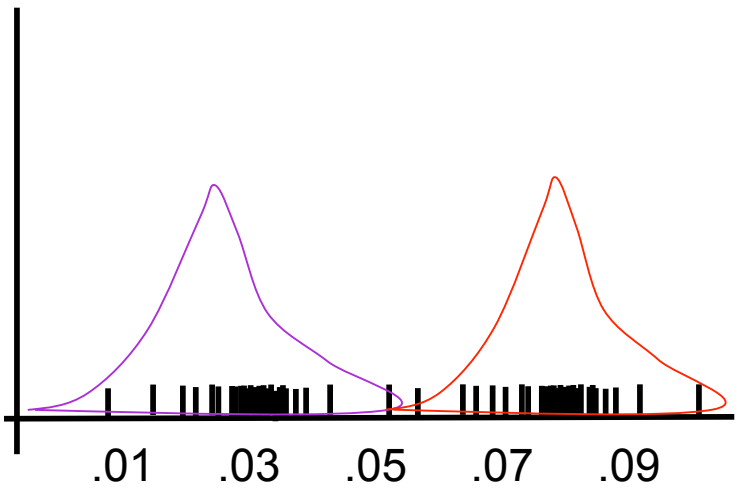
EM algorithm: Pictorial View



Simple example: learn means only!

Consider:

- 1D data
- Mixture of k=2 Gaussians
- Variances fixed to σ=1
- Distribution over classes is uniform
- Just need to estimate μ_1 and μ_2



$$\prod_{j=1}^{m}\sum_{k=1}^{K} P(x, Y_j = k) \propto \prod_{j=1}^{m}\sum_{k=1}^{K=2} \exp\left[-\frac{1}{2\sigma^2}\left\|x-\mu_k\right\|^2\right] P(Y_j = k)$$

EM for GMMs: only learning means

Iterate: On the t'th iteration let our estimates be

$$\lambda_t = \{ \mu_1^{(t)}, \mu_2^{(t)} \ldots \mu_K^{(t)} \}$$

E-step

Compute "expected" classes of all datapoints

$$P\left(Y_j = k \middle| x_j, \mu_1 \ldots \mu_K\right) \propto \exp\left(-\frac{1}{2\sigma^2}\left\|x_j - \mu_k\right\|^2\right) \mathrm{P}\left(Y_j = k\right)$$

M-step

Compute most likely new $\boldsymbol{\mu}$s given class expectations

$$\mu_k = \frac{\sum_{j=1}^{m} P\left(Y_j = k \middle| x_j\right) x_j}{\sum_{j=1}^{m} P\left(Y_j = k \middle| x_j\right)}$$

E.M. for General GMMs

$p_k^{(t)}$ is shorthand for estimate of $P(y=k)$ on t'th iteration

Iterate: On the *t*'th iteration let our estimates be

$$\lambda_t = \{ \mu_1^{(t)}, \mu_2^{(t)} \ldots \mu_K^{(t)}, \Sigma_1^{(t)}, \Sigma_2^{(t)} \ldots \Sigma_K^{(t)}, p_1^{(t)}, p_2^{(t)} \ldots p_K^{(t)} \}$$

E-step

Compute "expected" classes of all datapoints for each class

$$\mathrm{P}\left(Y_j = k \middle| x_j, \lambda_t\right) \propto p_k^{(t)} \mathrm{p}\left(x_j \middle| \mu_k^{(t)}, \Sigma_k^{(t)}\right)$$

Just evaluate a Gaussian at x_j

M-step

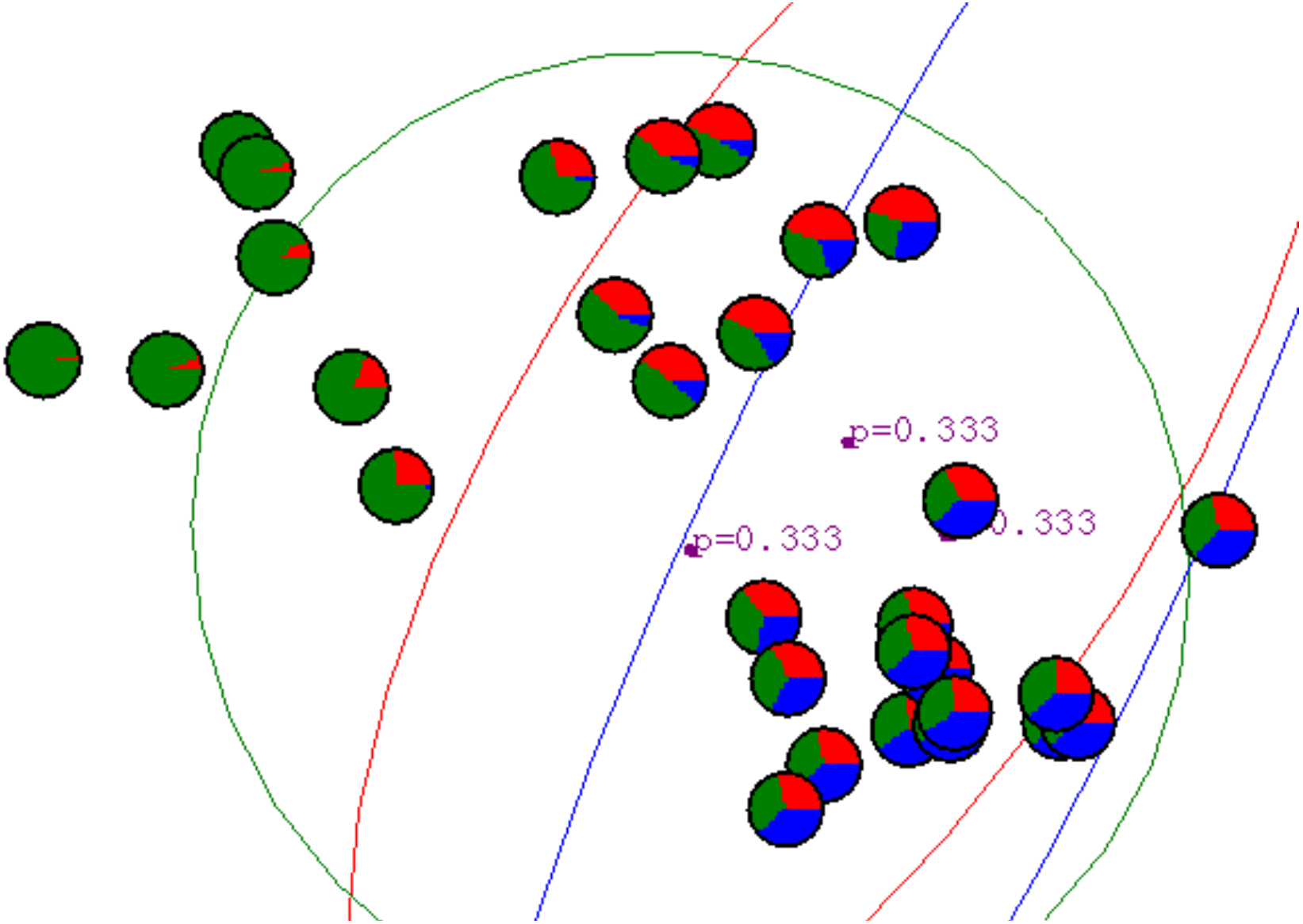
Compute weighted MLE for **μ** given expected classes above

$$\mu_k^{(t+1)} = \frac{\sum_j \mathrm{P}\left(Y_j = k \middle| x_j, \lambda_t\right) x_j}{\sum_j \mathrm{P}\left(Y_j = k \middle| x_j, \lambda_t\right)} \qquad \Sigma_k^{(t+1)} = \frac{\sum_j \mathrm{P}\left(Y_j = k \middle| x_j, \lambda_t\right) \left[x_j - \mu_k^{(t+1)}\right]\left[x_j - \mu_k^{(t+1)}\right]^T}{\sum_j \mathrm{P}\left(Y_j = k \middle| x_j, \lambda_t\right)}$$

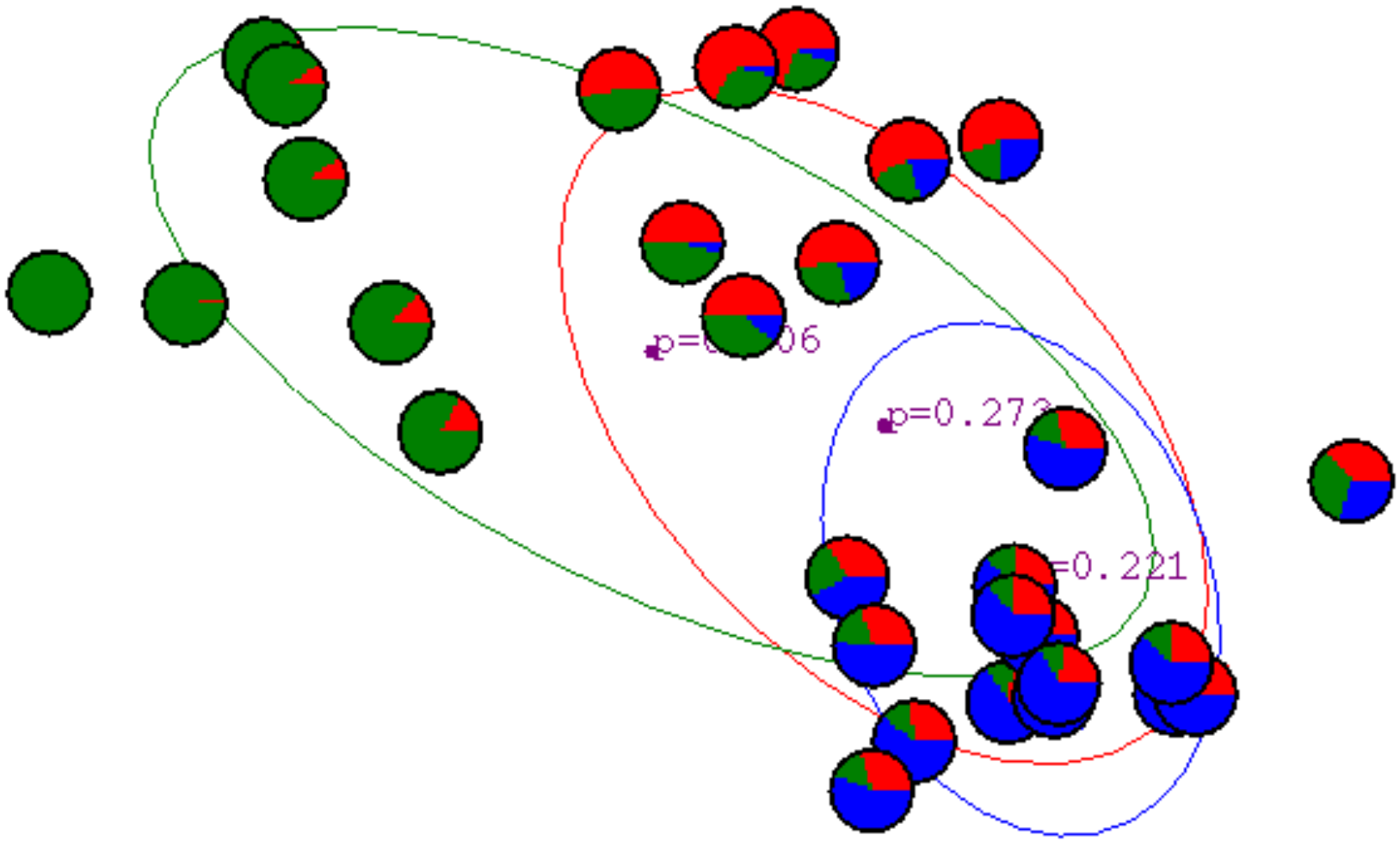
$$p_k^{(t+1)} = \frac{\sum_j \mathrm{P}\left(Y_j = k \middle| x_j, \lambda_t\right)}{m}$$

m = #training examples

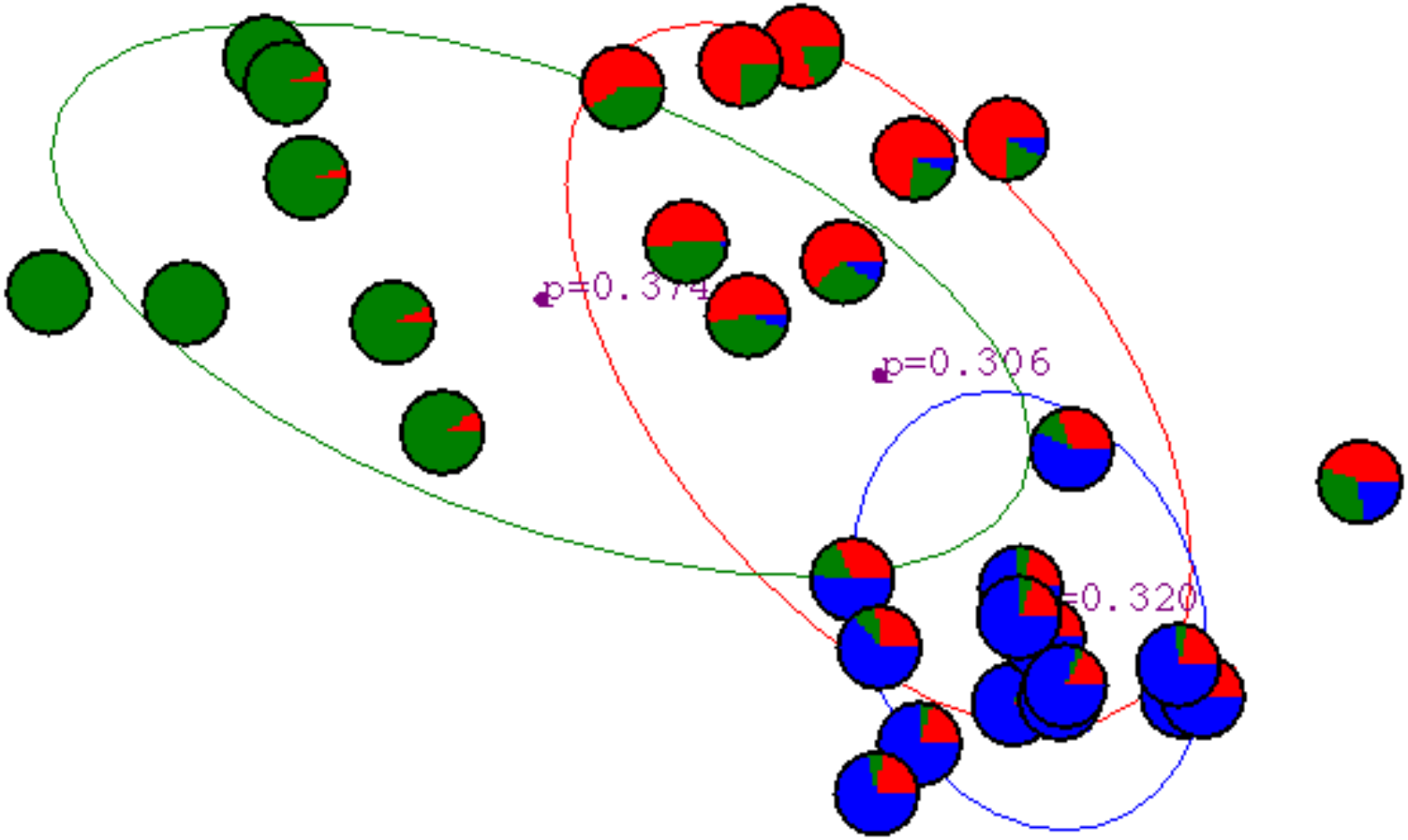
Gaussian Mixture Example: Start



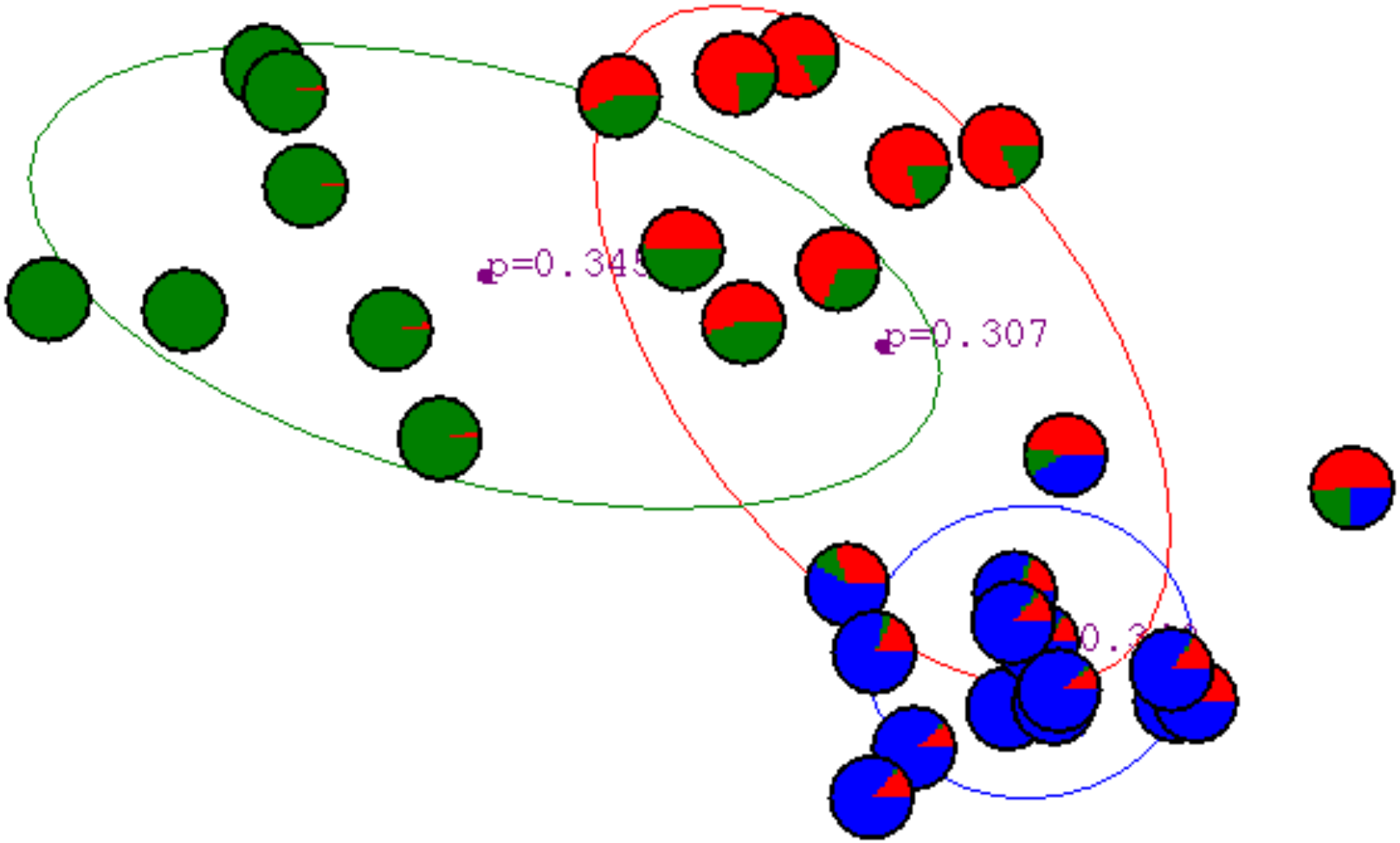
After First Iteration



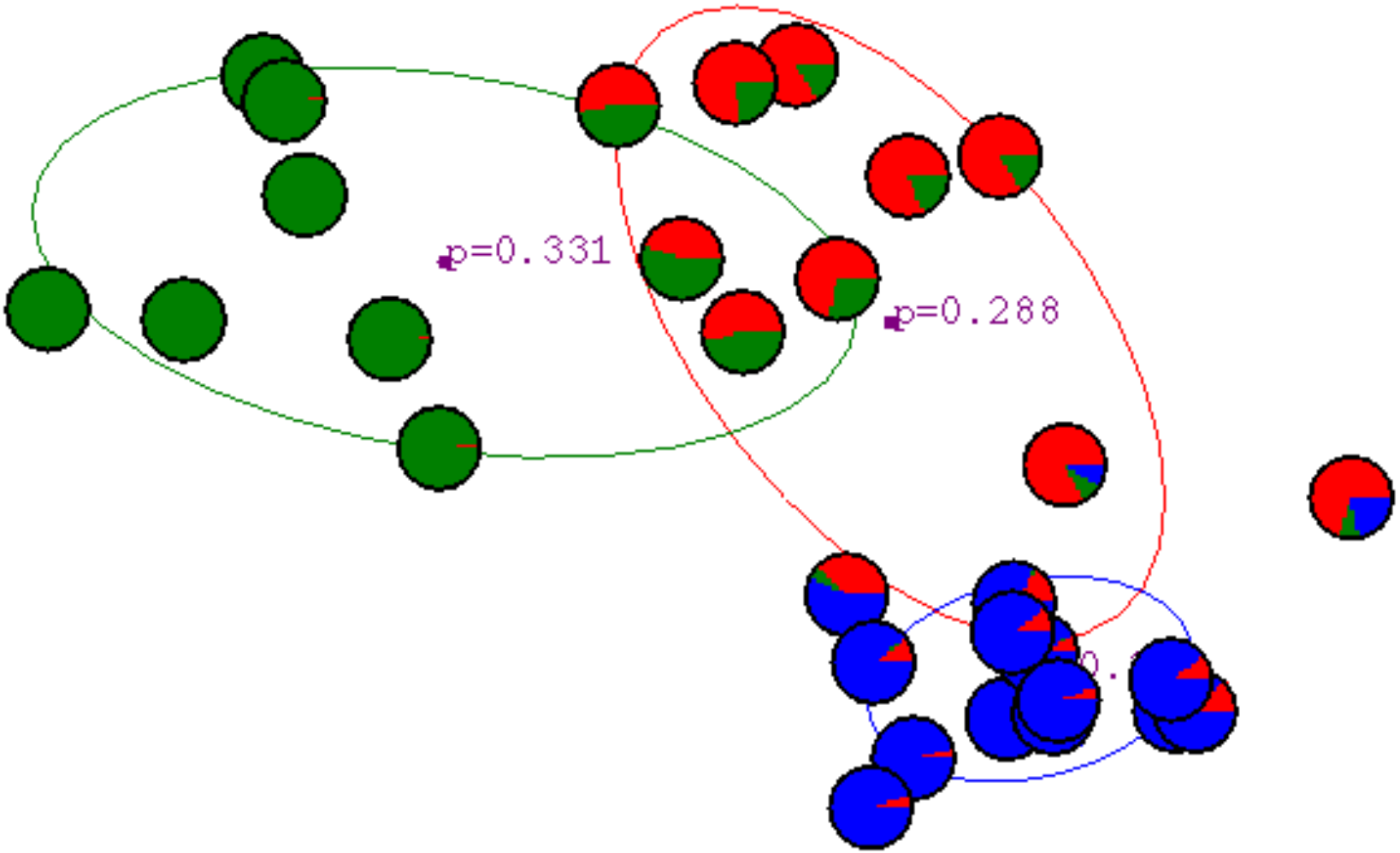
After 2nd Iteration



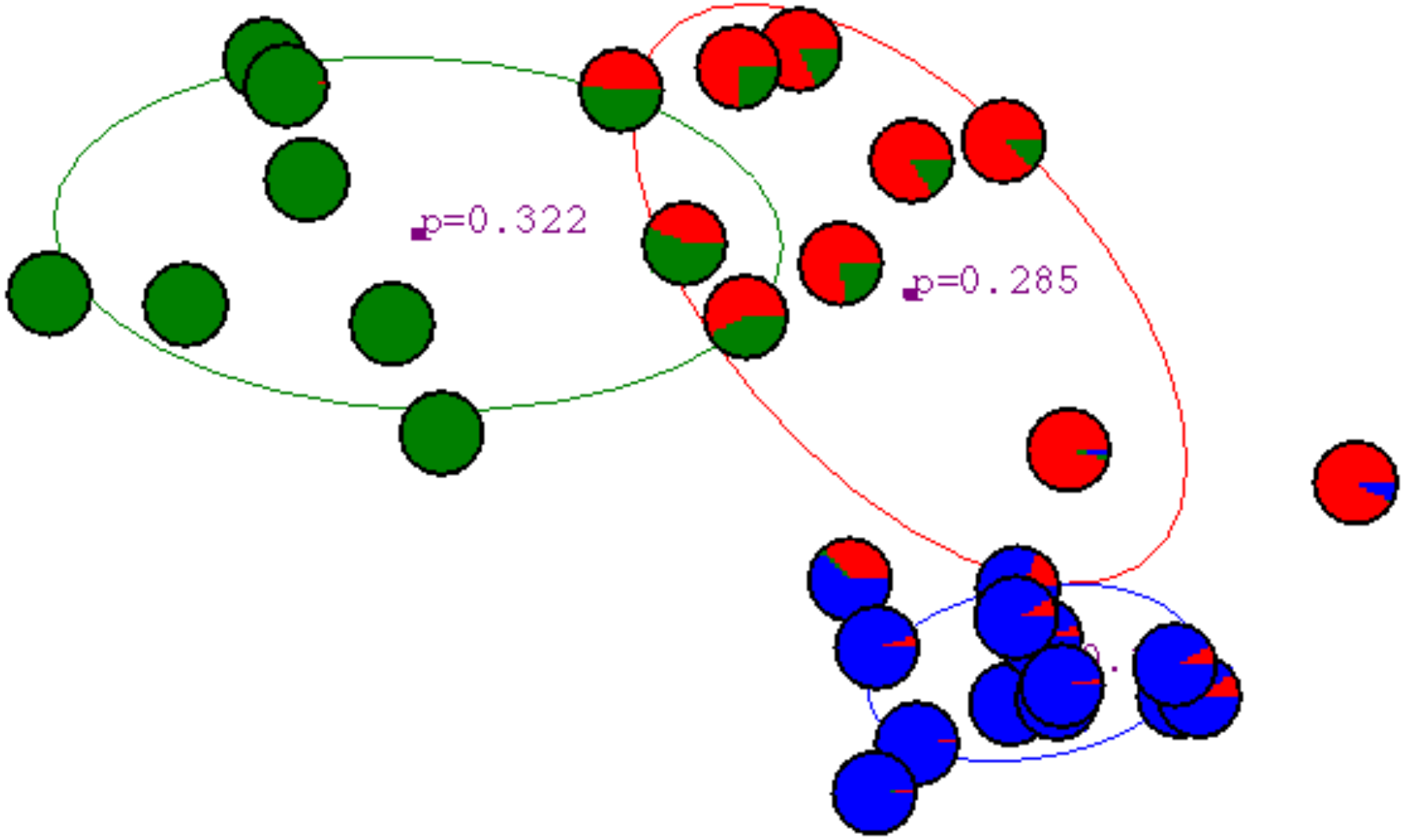
After 3rd iteration



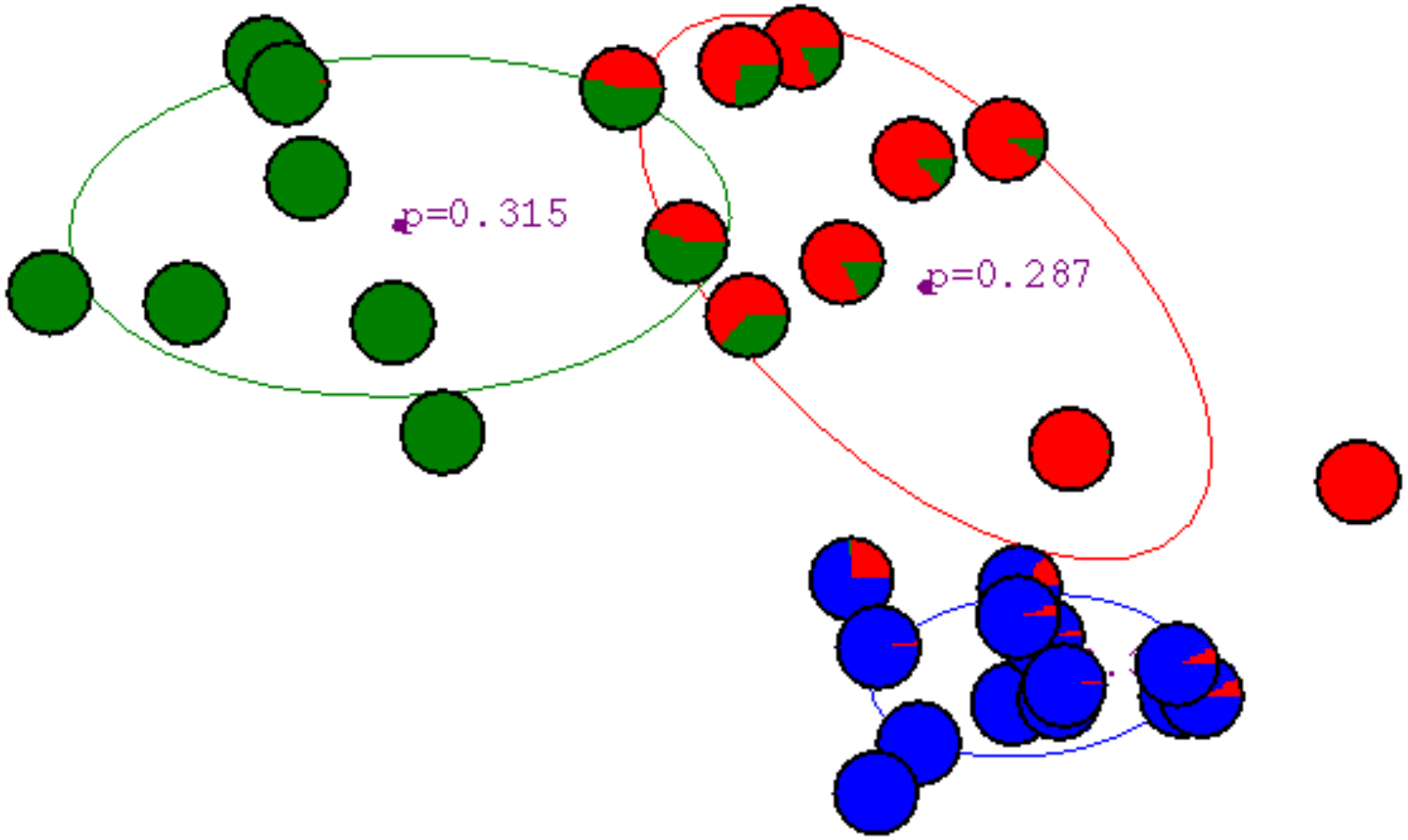
After 4th iteration



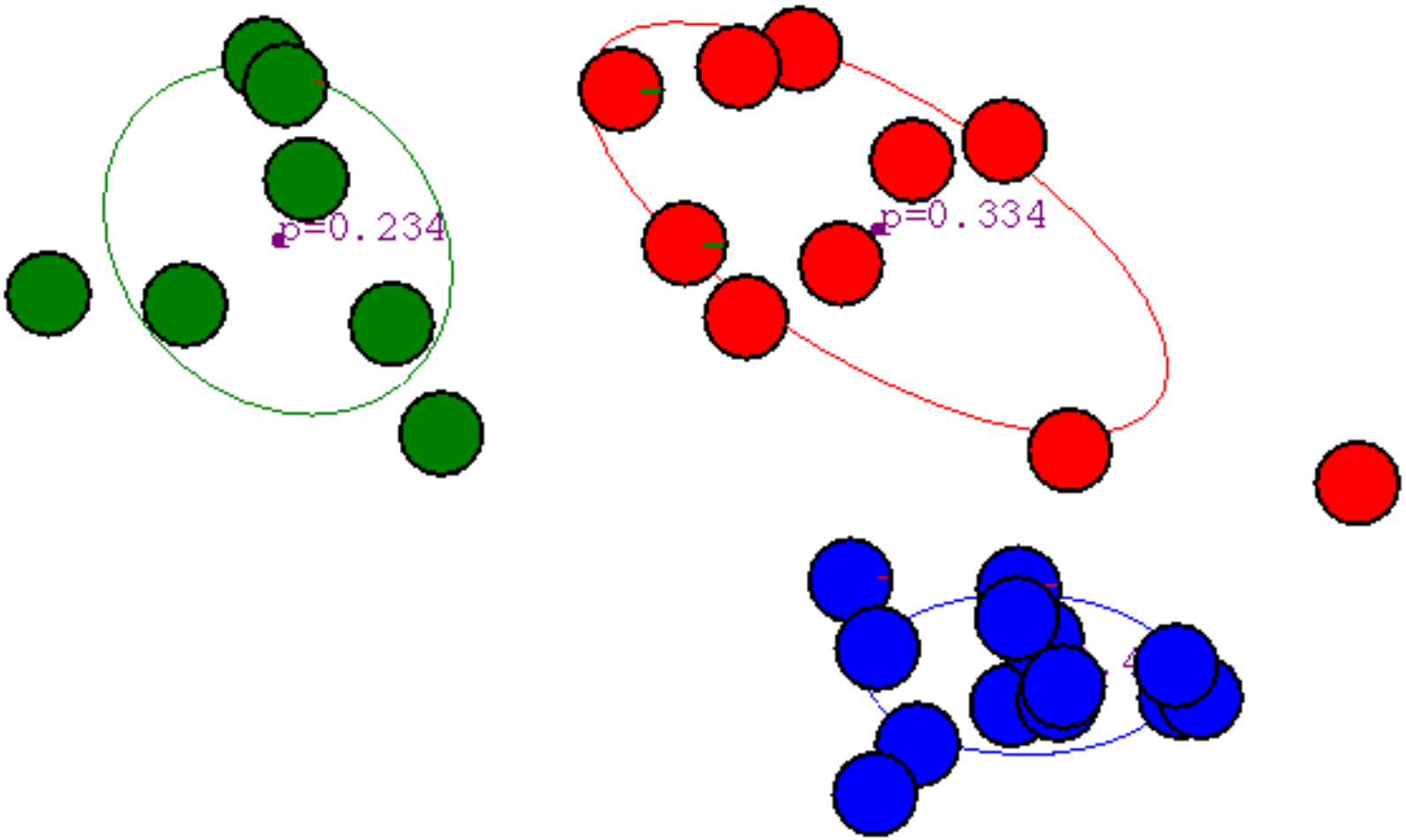
After 5th iteration



After 6th iteration



After 20th iteration



What if we do hard assignments?

Iterate: On the t'th iteration let our estimates be

$$\lambda_t = [\mu_1^{(t)}, \mu_2^{(t)}, \ldots \mu_3^{(t)}]$$

E-step

Compute "expected" classes of all datapoints

$$P\left(Y_j = k \middle| x_j, \mu_1 \ldots \mu_K\right) \propto \exp\left(-\frac{1}{2\sigma^2}\left\|x_j - \mu_k\right\|^2\right)\cancel{\mathrm{P}\left(Y_j = k\right)}$$

M-step

Compute most likely new **μ**s given class expectations

$$\mu_k = \cancel{\frac{\sum_{j=1}^{m} P\left(Y_j = k \middle| x_j\right) x_j}{\sum_{j=1}^{m} P\left(Y_j = k \middle| x_j\right)}}$$

$$\mu_k = \frac{\delta\left(Y_j = k, x_j\right) x_j}{\sum_{j=1}^{m} \delta\left(Y_j = k, x_j\right)}$$

δ represents hard assignment to "most likely" or nearest cluster

Equivalent to k-means clustering algorithm!!!

Implementation

`sklearn.mixture.GaussianMixture` implements GMMs within `sklearn`.

- `GaussianMixture` creates the class
  - `n_components` indicates the number of Gaussians to use.
  - `covariance_type` is type of covariance
    - `full`
    - `spherical`
    - `diag`
    - `tied` means all components share the same covariance matrix
  - `max_iter` is EM iterations to use
- Functions
  - `M.fit(X)` fits using the EM algorithm
  - `M.predict_proba(X)` is the posterior probability of each component given the data
  - `M.predict(X)` predict the class labels of each data point

Further Reading

- `sklearn.mixture.GaussianMixture`
- Python Data Science Handbook: GMMs